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On the management of social risks of hydraulic infrastructure projects in China: A case study

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Abstract

With social risk management attracting more attention in China, the Communist Party of China (CPC) Central Committee and the State Council released the "Guidance (trial) on Establishing a Sound Social Risk Assessment Mechanism". It is mandated that all infrastructure projects must pass social risk assessment prior to the project implementation. However, social risk management is in its infancy and has not formed a unified paradigm in China. In this paper, with an aim to explore how to manage social risks of infrastructure projects, particularly during the process of urbanization, a case study was undertaken on the identification of social risks based on an in-depth investigation of a hydraulic project. Related stakeholders were recognized in the first instance, followed by the assessment of social risks based on observations, expert meetings, interviews and discussion forums. Response plans were developed to prevent, mitigate and cope with the potential consequences of social risk events that may occur before or during the implementation process. The findings of this paper may provide a reference to the social risk management of future infrastructures.
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Keywords: Social risk; Infrastructure project; China; Case study

1. Introduction

Nowadays, there is a growing attention on social risks and associated management from both academics and practitioners (Branscomb, 2006; Jiang and Klein, 1999; Kwak and Stoddard, 2004). As a result, the risk management of projects is gradually addressed (Holzmann and Jørgensen, 2001). The risk society is a product of modern civilization, and is featured with globalization and equalization, which needs to be carefully dealt with (Beck, 1992). Till now, the study on the social risk of projects, particularly on the key infrastructures, spreads rapidly with diversified understanding in different countries and regions which addresses not only the macro political, social, and cultural issues but also micro project issues.

Since the 911 event in 2001, the United States of America has begun to place emphasis on critical infrastructures and key resources. The US National Infrastructure Protection Plan (NIPP) was established to "build a safer, more secure and more resilient America by preventing, deterring, neutralizing or mitigating the effects of deliberate efforts by terrorists to destroy, incapacitate, or exploit elements of Nation's Critical Infrastructure and Key Resources (CIKR) and to strengthen national preparedness, timely response and rapid recovery of CIKR in the event of an attack, natural disaster or other emergency" (Chertoff, 2009). Similarly, the United Kingdom and Canada attach great importance to risk management and the protection of infrastructure projects (CPNI, 2010; Public Safety Canada, 2010).

The last decades witnessed a large quantity of infrastructure projects invested and commissioned by governments in China. The extent and complexity of these infrastructure projects are astounding. The development of these infrastructures has

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significant social impacts "both because of the high level of efficiency they attain and the level of damage that a massive failure of those infrastructures might cause" (Branscomb, 2006). Without successful management, the construction of infrastructures often leads to the conflicts between local community and the project implementation organization, and subsequently leads to the cancelation or postponement of the projects. Even worse, community petitions and incidents originated by serious conflicts will affect social stability seriously (Xinhua Net, 2013). These include the cancelation of a chemical plant in Ningbo city, and a water pipeline project for a paper corporation in Nantong city as a result of large scale of public protest showing concerns on environmental issues associated with these projects (Ningbo government, 2012; Xinhua Net, 2012). This highlighted the critical role of social risk management on the implementation of infrastructure projects.

In order to mitigate and control the social risk of infrastructure projects during the urbanization process, some local governments in China start to require the social risk assessment of major infrastructure projects during the feasibility study stage such as Guiyang, Guizhou Province (Chen, 2009), Huangshan, Anhui Province (Huangshan Construction Committee, 2010), and Shanxi Province (Zhang et al., 2010). In 2012, the CPC Central Committee and State Council released the "Guidance (trial) on Establishing a Sound Social Risk Assessment Mechanism" (CPC Central Committee, 2012). This is an important milestone for the central government of China to place emphasis on social stability with an attempt to prevent social risk. It is mandated that no key infrastructure projects can proceed without passing the social risk assessment. The main purpose of social risk management is to achieve a harmonious society. The concept of social risk in Chinese context means the risks in any fields that may influence the whole society which may cause social turbulence and social unrest (Tong and Zhang, 2007); or may seriously affect society and people's productivity and quality of life (Murray and Grubesic, 2012).

To identify the social risks and deal with them correspondingly, a management system needs to be established. The function of such a social risk management system is to identify and subsequently mitigate, reduce and control those social risks which may occur before or during the process of project construction by means of robust risk response strategies and plans. In this paper, the authors attempt to examine how to establish a social risk management system of infrastructure projects via the case study of a hydraulic project during the process of urban renovation and renewal in the city of Ningbo, Zhejiang Province of China, and how to manage those risks in a proper manner.

2. Literature review

Holzmann and Jørgensen (2001) defined social risk management as "… public interventions to (i) assist individuals, households, and communities better manage risk, and (ii) provide support to the critically poor" (p. 530). They consequently suggested that three main strategies to manage social risks, i.e. prevention, mitigation and coping strategies depend on when the strategy is introduced from the occurrence of the social risk event. In essence, social risk management places focus on the social outcome or social objective of business activities. In project management context, social risk is closely linked to stakeholder management with not only the traditional player such as contractors and employees but also the public and the community (Aaltonen, 2011; Aaltonen et al., 2008; de Bakker et al., 2011). Such risk management exercise also features corporate social responsibility in order to deal with environmental and social impacts of business activities (Kytle and Ruggie, 2005; Zhao et al., 2012). Indeed, Kasperson et al. (1988) pointed out that risks could be socially amplified from direct to indirect consequences due to behavioral responses spread to other stakeholder groups. Benefits of managing social risks properly include: reducing vulnerability, enhancing smooth consumption, improving equity, smoothing household welfare, and reducing poverty (Holzmann and Jørgensen, 2001).

In the construction sector, the social risk was classified by Bing et al. (2005) as a type of macro level risks with factors such as "lack of tradition of private provision of public services" and "level of public opposition to project". It is well recognized that social risk is one of the most common risks that exist in infrastructure projects that correlates to other risks such as physical risks and subcontractor related risks which should not be overlooked (de Lemos et al., 2004; Ghosh and Jintanapakanont, 2004; Gilmour et al., 2010). Miller and Lessard (2001) pointed out that social acceptance was one of major institutional risks in large infrastructure projects. The level of social acceptance of infrastructure projects depends on how the social stakeholder groups' benefits and impacts are influenced by the projects from a long term perspective (Yuan et al., 2011). According to Zavadskas et al. (2010), social risks are normally overlooked by project stakeholders however with significant impacts on project outcomes. Bredillet (2008) suggests that social risk assessment and mapping forms a critical component of project management governance. There are both external and internal risks associated with infrastructure projects. The internal risks usually include labor demands, financial problems, quality and safety issues, which can be controlled through successful project management. The external risks, however, such as unexpected changes of policy, and nature disasters, are hard to be controlled (Lee et al., 2009; Lyons and Skitmore, 2004). Social risk, highly related to social stability, is also one of these external risks. The indicators of social stability include public health, emergency services, information and telecommunications, transportation (people and product), banking and finance, etc. (US Office of homeland security, 2002). Following this stream the social risk included terrorism, catastrophic natural disasters and catastrophic accidents as well as other hazards (US Office of homeland security, 2007).

However, since the management of social risk of major infrastructure projects hasn't form a unified standard in China, the understandings are diversified (Chen, 2010; Jiangxi Development and Reform Committee, 2011). For infrastructures in China, Liu et al. (2005) classified the social risks into four types, i.e., value risk, organization risk, information risk and environmental risk. While the Jiangxi Development and Reform Committee (2011)

of Jiangxi province proposed the index of assessment system from another perspective, which included economic, ecology environment, social policy and system. In addition, Dong (2011) suggested the index to include legality, legitimacy, feasibility, safety and controllability. These studies addressed some aspects of social risk management on infrastructure projects. However, there is still a lack of effort to investigate how to assess and mitigate social risks at the tactical level.

Besides establishing an index system, it is also imperative to prioritize the indicators. A two-dimensional method has been commonly utilized to describe project risks, including the probability of risk events (p), and its potential loss (q). The level of risk (R) is a function of p and q, i.e., R = f (p, q) (Al-bahar and Crandall, 1990). Williams (1996) critiqued that this method missed some critical information of risks which might lead to inaccurate result. He suggested that the predictability, arbitrariness and cognitive of risk should also be taken into account. In addition, expert evaluation approaches such as AHP combined with fuzzy reasoning were utilized to assist the quantification and prioritization of risk factors (Khazaeni et al., 2012; Nieto-Morote and Ruz-Vila, 2011; Zeng et al., 2007). Similarly, it is imperative to assess risks both qualitatively and quantitatively with inputs from experts (Shi et al., 2014). A typical example is that Wang (2011) analyzed the social risk of a fire engine access project through AHP which found that the acquisition was the main risk which may cause dissatisfaction of the local community. This social risk could be avoided by means of improving the acquisition policy. In the social risk assessment of a main road project in the urban area, Wang divided the social risks into three categories, i.e. environment, policy and project management and consequently assessed the risks via the two-dimensional description method. So far the cases of social risk assessment in China are still very limited. In addition, no common sense exists on which method is most suitable for social risk management. Moreover, previous studies overlooked some new opportunities for the city and facilitation of urbanization from renewal developments. In this paper, the Authors attempt to examine how to systematically manage social risks of infrastructure projects through the case study of the Yanshan Trunk River Renovation project.

3. Methodology

3.1. Case selection

In the beginning of this research, there were three candidate projects to be selected as the potential case for study. These were the "three Gorges project" in China, which was the largest and the most well known hydraulic project in China, the Costal Industry Park in Nantong City, which was a typical project for new town development, and the river renovation in the city of Ningbo of Zhejiang Province.

"Transparent observable" is important for case selection (Pettigrew, 1990; Eisenhardt, 1989). This is also the most critical consideration of case selection in this study. Following this major premise, four criteria were set up for case selection in order to examine the research question on how to establish a social risk management system of infrastructure projects within the context of urbanization and how to manage those project risks correspondingly. These four criteria are:

(1) It must be a typical infrastructure project;
(2) It needs to be related to urbanization;
(3) It should face to tough social risks during project development and construction;
(4) It must be an ongoing project to allow tracking the whole process of social risk management of the project.

As a result, the Yanshan Trunk River Renovation Project was finally chosen as the case for this study. The reason of the selection is as below.

Firstly, the Yanshan Trunk River Renovation project is a hydraulic infrastructure project. Large hydraulic infrastructure projects have been an important component of infrastructure development in both capitalist and communist countries (Jackson and Sleigh, 2000). The catastrophic flooding often caused enormous economic damage and casualties. Hydraulic infrastructure projects can not only prevent flood effectively but also bring some economic benefits (Jonkman et al., 2008). Therefore, hydraulic projects, which are with strong public welfare, fundamental and strategic have become an irreplaceable foundation support for economic and social development. In China, the distribution of water resource is uneven. Government has vowed to deal with this issue by means of developing hydraulic infrastructures to prevent flood and drought. A large number of hydraulic infrastructure projects were constructed in recent years. In Zhejing Province, during the period of Twelfth Five Year Plan, many key rivers need to be regulated. The Yanshan Trunk River is one of them. These projects can bring many advantages. For the water environment, the hydraulic projects can regulate mesoclimate and improve landscape. For social environment, it can improve the capability of preventing flood and improve the water quality. Similarly, it can attract tourists and improve irrigation so as to facilitate the development of local economy (Wang et al., 2008). On the other hand, hydraulic infrastructure projects usually face social problems as well (Jackson and Sleigh, 2000), which calls for the research on the social risks.

Secondly, the Yanshan Trunk River Renovation project is more typical in this study than the Three Gorges project. It is more related to the city renewal and involves typical social risks in the process of urbanization in China, such as land acquisition and housing demolition, produce loss of enterprises, and transport impact. Three Gorges project is a large dam-building project where the main issue is migration. By contrast, Yanshan Trunk River Renovation includes not only the function of engineering projects but also the other ones such as improving the urban image and driving the local economy.

Lastly, the Yanshan Trunk River Renovation project conforms to the fourth criterion. This project began in 2009 and will be completed in 2014. There exists plenty of information of social risk management that is useful for this study. In contrast, the Nantong Costal Industry Park just commenced, lacking of relevant data.

3.2. Case study protocol

Following the selection of the case project of Yanshan Trunk River Renovation, a case study approach was designed for this study. Case study approach is most suitable for how and why type of research questions (Yin, 2003). The advantages of case study approach include the ability to incorporate multiple sources of information and to highlight the success or challenges presented to a project (Barrett and Sutrisna, 2009; Herazo et al., 2012; Müller et al., 2013). As a result, case study approach has been applied in numerous project management studies. Indeed, Yin (2003) pointed out that the case study was an effective approach to understand the complex phenomena. Since the research on social risk management of infrastructure projects is in its infancy in China, the case study method is suitable for establishing primary theory through observation, expert meetings, discussion forms, and interviews of related stakeholders of the case project.

Firstly, research questions were designed on how to identify and assess the social risks on infrastructure projects within the context of urbanization and how to manage those risks correspondingly. The questions were further developed into three focused sub-questions, i.e. (1) What are the main factors that affect social risks for infrastructure projects within the context of urbanization? Why? (2) How to assess those risks? (3) How to develop the response plan to deal with those risks? Afterwards, those stakeholders related to the social risks of the case project were interviewed together with organizing several meetings during the initialization stage of the projects in order to identify the potential social risks. The interviewed stakeholders include: (1) the local government leaders, including the ministers of construction department, financial department, land planning department and governmental office; (2) the representatives of enterprise owners, including the principal of a hat-making factory, a garment enterprise, a recycling company, a textile enterprise, and a chemistry industry factory; and (3) the representatives of villagers, including the heads of 20 villages. Afterwards, the identified risks were quantified and the response plans were developed accordingly based on an experts' evaluation. Subsequently, the progress of the project was monitored to ensure that the social risks were under control. Finally, a seminar was organized to convey research findings to related stakeholder groups. The total period of this in-depth case study is 17 months accompanied with the progress of Yanshan Trunk River Renovation project.

4. Case of Yanshan Trunk River

4.1. Background

4.1.1. Overview of the project

This infrastructure project was to build the dams along the coast of Yanshan Trunk River, which was located in the southeast plain of Ningbo, Zhejiang province. It was one of the three major backbone drainage systems, and was an important project of "Ningbo Urban River Renovation Plan" and "Southeast Ningbo Flood Control and Drainage Plan" (Zhejiang Province Government, 2004). The aim of the project was not only to prevent flood, but also to improve the landscape and the living environment of the inhabitants on both sides of the river. In addition, the project was expected to attract more tourists.

The dams covered through 5 boroughs including 20 administrative villages from south to north. The total length of the regulated river to be renovated was 29.7 km, including the water diversion tunnel project of the upper stream, the broaden work of the watercourse of the trunk river of 25.6 km, the water gate of the downstream, the regulating sluice, the reform work of 22 crossing bridges, revetment afforest and the roads on both river banks.

The project needed 8377 acres of permanent land acquisition and 136,200 m^2 of housing demolition. The construction period was 60 months. The work elements of this infrastructure project are shown in Table 1.

4.1.2. The stakeholder groups of land acquisition and housing demolition

According to the different types of the land that involved the acquisition and demolition, there were three related stakeholder groups of the project. The analysis of these stakeholder groups and their demands are shown below.

(1) Villagers affected by land acquisition

This project needed to occupy some lands which planted economic crops; and some fishponds. As a result, the land acquisition process would bring losses to the peasant households. Similarly, the construction of the watercourse would influence the irrigation system. Therefore, it was imperative to define the reasonable compensation for these villagers affected by the land acquisition process. This would directly influence their quality of life and reduce the associated social risk. The villagers generally had three appeals:

Firstly, the compensation standard was the focal issue that the villagers cared most. The villagers hoped that the compensation standard could be defined as soon as possible and must be fair and transparent. The standard should conform to the laws and regulation and take the immediate and long-term interests of villagers into account. Equity was another critical issue with special attention to be paid to disadvantaged groups.

Disclosing the compensation standard and related policies helped to stabilize the public mood and to clarify misunderstandings so that the land acquisition could be undertaken more effectively. During the study, the villagers had expressed urgency to understand the compensation standard of the loss.

Secondly, villagers demanded the land acquisition should be undertaken as soon as possible once the compensation standard was defined. On the one hand, it could minimize the inconvenience of the rural production caused by the land acquisition. On the other hand, it could also avoid unfair events to some extent.

Thirdly, efforts were required to reduce the impacts on the cultivation and irrigation of other farmlands.

Table 1
Work elements of the Yanshan Trunk River Renovation Project.

No.	Content	Unit	Quantities	Note
1	Hydrology			
1.1	Drainage area	km^2	703	Plain: 403 km^2 mountainous areas: 300 km^2
1.2	Average annual precipitation	mm	1510	
1.3	3-day maximum rainstorm in once-in-twenty years	mm	345	
2	Project scale			
2.1	The length of regulated river (to be renovated)	km	29.7	
3	Land occupation			
3.1	Permanent	acre	8377	
3.2	Temporary	acre	3909	
4	Key construction			
4.1	Upper stream			
4.1.1	Check sluice	Number × m	4 × 4	
4.1.2	Tunnel	m	1408	Clear width: 5 m, height: 7.5 m
	Tunnel entrance/exit height	m/m	0.5/−1.87	
4.1.3	Length and width of canal for irrigation	m/m	333/20–9.48	
	Height of canal for irrigation	m	1.0–0.5	
4.1.4	Length and width of downstream exit section	m/m	66/7–30	
4.1.5	Length and width of downstream transition section	m/m	116/30	
4.1.6	Length and width of downstream connecting section	m/m	2166/30	
	Height of downstream connecting section	m	−1.75	
4.1.7	Bridges involved in abstraction works of upstream	Number	5	33 (length) × 10 (width)
4.2	The trunk of the river	km	25.6	New excavated river: 850 m
4.2.1	Width	m	40–60	
4.2.2	Bottom depth	m	−0.87 to −1.87	
4.3	Water gate	Number × m	7 × 5	
4.3.1	Gate depth	m	−2.0	
4.3.2	Foundation feature		Sludge (sludge clay)	
4.3.3	Gate		Reinforced concrete Beam-slabplat gate	
4.4	Pump station	m^3/s	100	8 + 1pumps/12.5 m Type: 3s1800GZ-160
4.5	Bridge over the river	Number	22	

Interviewees showed their concerns that the construction of the watercourse may affect the irrigation system severely. Therefore, the villagers expressed their demands that the construction works should minimize the influence to other irrigative lands and the irrigation system should be rebuilt for the farmers.

(2) Villagers affected by housing demolition

The project involved a certain amount of housing demolition along both sides of the river. The demolition needed to be open, fair and just. The government should take care of the benefits of the vulnerable groups and dispose the demolition and resettlement work so as to reduce related social risks. These villagers had three appeals during the housing demolition:

The first and second appeals were similar to those of villagers affected by the land acquisition. They also concerned about the compensation standard and urged the housing demolition could be undertaken as soon as possible with attention paid to security of the residents and their belongings. The third appeal was to dispose the illegal structures in a proper manner. The villagers demanded that the housing demolition process should guarantee the benefits of project and government with due regard to villages' interest.

(3) Resettlement enterprises

The project also involved the relocation of some enterprises, including a hat-making factory, a garment enterprise, a resource recovery company, a textile enterprise, and a chemistry industry factory. As the land acquisition and demolition work would occupy some of lands originally occupied by these enterprises, the production mode and contents would be affected to a certain degree which led to the loss of their income. Therefore, the government should provide the enterprises reasonable compensation for the operation loss and mitigate related social risks. The enterprises had three appeals:

Firstly, the enterprises demanded reasonable compensation for the relocation and operation loss. To most enterprises, the acquisition of land would affect their production. As a result, these enterprises would not be able to deliver the order in time. Therefore, the enterprises urged the government to take this into account, and raised the compensation standard so as to make up for the loss of both relocation and operation.

Secondly, the enterprises required that the resettle plot could be determined as soon as possible. The land was the main place of production for these enterprises and it was the main source of income. The land acquisition, to a

certain degree, would affect the production and future development of these enterprises. Therefore, for the long-term interests, in order to maintain normal production, owners of these enterprises generally preferred the government to provide reasonable resettlement land, rather than the monetary compensation. The government must handle this demand properly as it will not only affect the enterprise's benefit, but also influence the local economic development and employment opportunities.

Thirdly, the government should develop the resettlement schedule and scope reasonably in order to reduce the cost. Some of the enterprise owners required that the construction should be in accordance with the land conditions. Similarly, they hoped that the construction plan could be tailored to different enterprises and reserve some lands for production. They suggested resettling the factories on one side of the river at one time so as to reduce the cost.

4.2. Scope and approach of risk management

4.2.1. The scope of risk management

According to the investment administration system and project construction procedures mandated by the Chinese government, the project was divided into four stages, i.e. pre-preparation, preparation, implementation and operation. The risks affecting social stability existed in all four stages of the project and each has its own characteristics. If not being handled properly, many social risk events may be triggered such as blocking transport and destroying construction site that affect social stability.

Based on the four stages of the project, the scope and contents of risk management are as follows:

(1) Risks during the early stage of the project. This phase was from the initialization of project planning to the government approval of the feasibility study report. The social risks during this stage mainly related to the policy, planning, land, environmental protection, resource, preliminary construction plan, land acquisition and demolition plan, social conditions and public opinions, orientation of social media, etc.
(2) Risks during the preparation stage of the project. This stage was from the government approval of the feasibility study report to the official commencement of the project. The social risks during this stage were mainly related to policy, financing, bidding, design scheme, resettlement plan of land acquisition and demolition, social conditions and public opinions, orientation of social media, etc.
(3) Risks during the construction stage of the project. This stage was from the commencement to the completion and commissioning of the project. The risks included technologies utilized during the construction phase (e.g. building, structure, equipment, and construction), safety risk (e.g. construction site, labors, and material handling), environmental risks (e.g. construction and demolition waste, dust, sewage, exhaust gas, and noise), transport risks, construction schedule management, etc.
(4) Risks during the operation stage of the project. The risks were mainly related to the functionality of the project; and the impacts of project operation on surrounding communities, environment, transport and safety, etc.
(5) Other risks that involved interests of the local community.

4.2.2. The approach of risk management

Prior to the project approval, the government established a social risk management group of the project, including related departments, experts and scholars or consultants, to assess the social risks and to develop measures of safeguarding social stability accordingly.

As the members of the social risk management group, the research team reviewed the proposal files, the environmental impact evaluation report, the feasibility report, and other related project documents. Subsequently, they hold a meeting with the key project participants to collection related information about project progress. During the progress of the risk assessment, the research team visited the towns and villages repeatedly to interview the villagers, recognized the stakeholders and investigated their attitudes to the project. At the same time, the stakeholders such as the householders related to the resettlement were invited to a discussion forum on the compensation scheme. Moreover, related departments were consulted about the petition and the social risk of other infrastructure projects currently undergoing. Based on all these inputs, the social risk factors were analyzed in Section 4.3. Afterwards, experts were approached to demonstrate and assess the social risk of the project in Section 4.4 and attempt to develop the response plan in Section 4.5. The effectiveness of the response plan was monitored regularly and reported back to the management group for continuous improvement. The system of risk management is shown in Fig. 1.

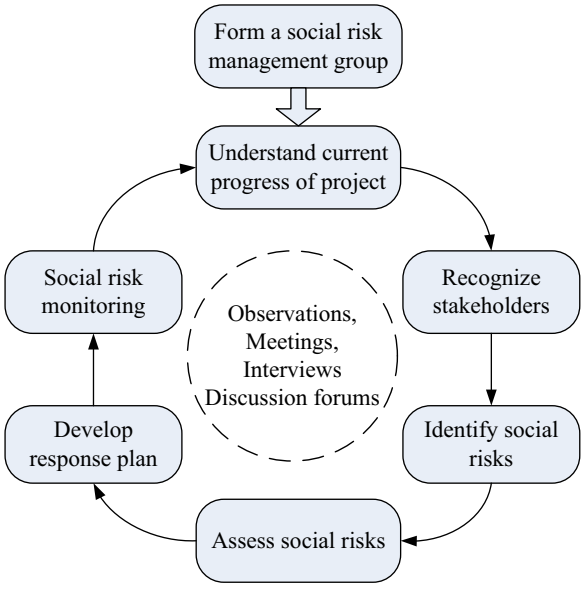


Fig. 1. The system of social risk management.

4.3. Social risk identification

To manage the social risks of the infrastructure project effectively, the factors of the social risks associated with the project were identified. Consequently, these social risk factors were analyzed from two aspects, i.e. the likelihood of the occurrence of each risk and the severity of the consequences if the risk occurred based on the observations, meetings, discussion forums and interviews related stakeholder groups. These factors are shown in Fig. 2.

4.3.1. The legality and the rationality of the project

The factors related to the legality and the rationality of the project included: whether the project conflicted with policies, laws or regulations; whether it had relevant policies and legal basis; whether the project adhered to strict investigation and approval procedures; whether the project went through rigorous scientific feasibility analysis with a thorough consideration of constraints such as time, space, human resources, financial, and material; and whether the imposed plans were feasible. All of these factors were prerequisites for the support and assistance from all levels of government, organizations and the general public which was critical for the successful implementation of the project.

This study found that all related stakeholder groups had no doubts on the legality, rationality and the necessity of the project. The project went through sufficient feasibility study and was in strict accordance with the Regulation on the Expropriation of Buildings on State-owned Land and Compensation. At the same time, the project complied with the general plan for land utilization, town and country plan, and the specialized plan of Ningbo city. It was also beneficial to the regional development.

The project would drive the infrastructure development in the local area such as new construction of riverbank and

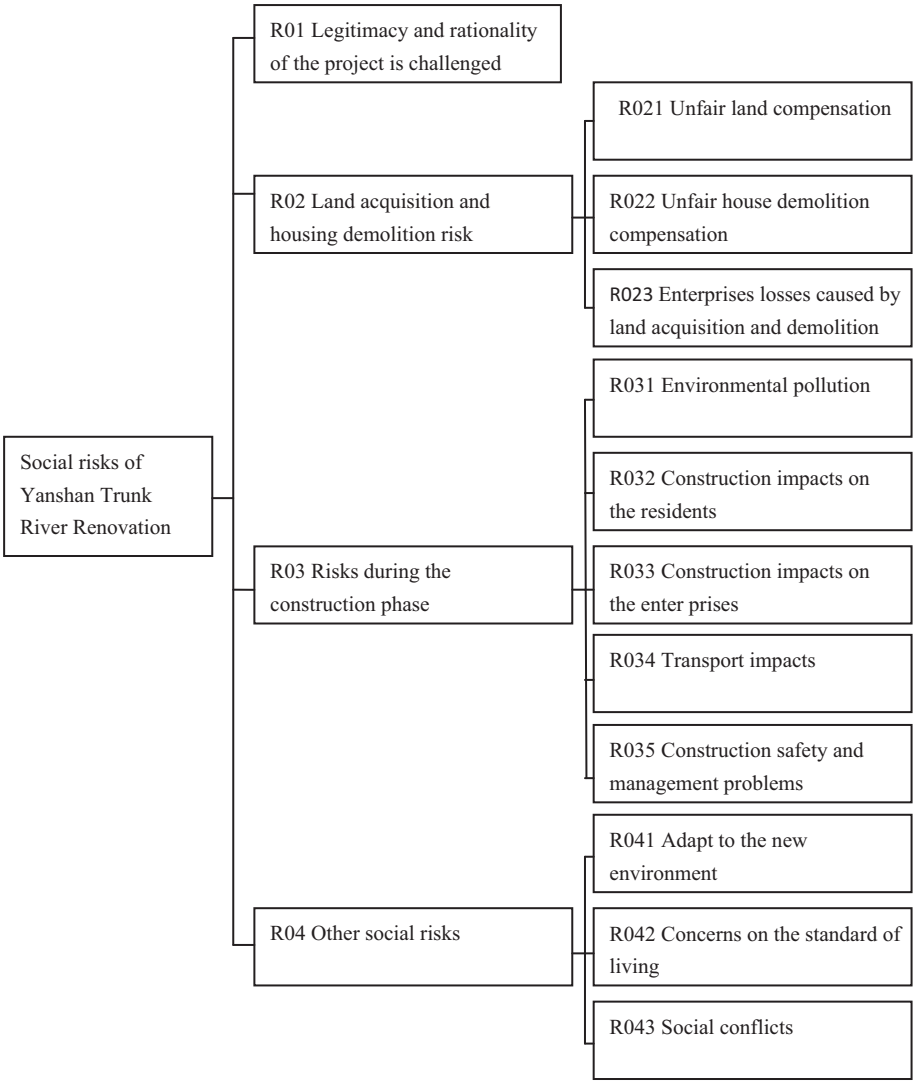


Fig. 2. Social risks of Yanshan Trunk River Renovation Project.

bridges. Apart from flood control, the project would improve the landscape, and motivate the development of cultural tourism industry, which was beneficial to the local economy.

However, efforts are required to ensure the legality and the rationality in other infrastructure projects as well due to their large scale and significant impacts on the society and environment. If any illegal operation exists, the local government would lose the trust of the villagers which may lead to severe social risks.

4.3.2. Land acquisition and housing demolition risk

The land acquisition and demolition involved the immediate interests of the villagers, some of which did not understand the policy. As a result, social risks existed if the government did not handle this issue properly.

During the progress of the land acquisition and demolition, the related stakeholder groups paid more attention to reasonable compensation. Moreover, the land acquisition and demolition could bring varying degrees of influence to the quality of life of villagers and the production of enterprises. Their major income source may be reduced due to the implementation of the project. Therefore, a reasonable compensation needed to be provided in order to make up for their losses.

Despite showing a general understanding of the necessity of the project, related stakeholder groups still required fair compensation. Otherwise social incidents may occur with severe impacts. As the land acquisition and demolition would bring loss to both villagers and enterprises, it was imperative to understand what they expected for compensation. If the compensation could not meet their expectation, the residents may boycott resettlement and delay the construction schedule. This presented significant social risks.

The probability of the risk was high which most likely occurred in enterprises. First, the owners of the enterprises preferred land replacement than monetary compensation. However, the stock of the land, which was available and fit the requirements of the enterprise, was not very large. Therefore, the probability of land replacement was small, which may cause the rejection or resistance of the enterprises. Moreover, the owners of these enterprises wanted to maintain the normal production of the factories, which needed an ideal construction scheme that took this into consideration. This presented significant challenges to the project management team.

For these reasons, the likelihood of land acquisition and housing demolition risk was high and the consequence of this risk was severe which may lead to not only the loss of the villagers and enterprises, but also the negative impacts on the economic development of the region.

4.3.3. Risks during the construction phase

During the construction phase, the villagers and enterprises outside of the project boundary may be affected as well. These included ecology impacts of construction activities, traffic congestion, and impacts on the production, which may lead to risks of social instability.

(1) The ecological impacts

The ecological impacts during the construction phase included noise, water environment, air environment and other environmental impacts.

According to the scale, site layout and mechanical equipment of the construction, the sources of noise during the construction phase included traffic noise, machinery noise and blasting noise. According to the environmental evaluation report, the possible traffic and machinery noise would be 75–90 dB, and the blasting noise of the tunnel would be 105–110 dB, which had potential impacts on daily life of local residents and production of enterprises in the region during the daytime.

Moreover, as the project is close to the watercourse, the construction may affect the water quality. According to the environmental evaluation report, during the construction period, the discharge of sanitary sewage would be about 17,000 m^3, average daily 15 m^3/d, and the concentration of COD_{Cx}, BOD_5, ammonia nitrogen, and SS would be 300 mg/l, 200 mg/l, 35 mg/l, and 200 mg/l respectively. Besides sanitary sewage, the construction machinery, such as swamp bulldozers, loaders and transport vehicles, would also produce wastewater, which would reach 3 m^3/d, including some oil pollutants. The concentration of SS would reach 2000 mg/l. Moreover, the construction needed 359,500 m^3 of concrete and 62.9 m^3 of building blocks, which would produce a large amount of wastewater. During the late period of construction, the concentration of suspended solids would be very high, which may reach 400–20,000 mg/l. If these pollutants were discharged into downstream directly, the water quality would be influenced significantly and caused sedimentation.

Similarly, the dust and exhaust gas from the construction activities affected the air quality of the construction site and peripheral villagers. The dust and gas often came from the construction machinery, the vehicle, the stir of the concrete, and blast works. The observation and investigation of the construction site showed that the dust caused by raw material transportation and tunnel excavation was the main pollutant. Moreover, construction vehicles produced a large quantity of exhaust gas, such as NO_2, CO_2 and some foul gas, such as hydrogen nitride and hydrogen sulfide when clearing the watercourse.

Last but not the least, construction activities introduced changes to the ecosystem integrity, including the original topography and land use. This may bring potential impacts on ecological balance, and even the whole ecosystem. Due to the site clearing works such as excavation, filling and grinding the earth-rock, the original topography and vegetation may be destroyed which led to the bare ground. Similarly, the change of the topsoil structure weakened the soil's ability against corrosion, and increased soil erosion of surface runoff, which may bring negative effect on river drainage and agricultural practices.

As the house and cultivated land of villagers were far from the construction site, the probability of this risk affecting villagers' daily life was not very high.

However, the probability of negative impacts of construction activities on environment, particularly on water resources, was high.

(2) The transport impacts
The construction site of the project was located at the border of the villages. Therefore, the traffic condition was not very busy or complicated. As a result, the extent of the traffic impact was not very large. During the initiation period of construction phase, some minor traffic issues occurred as the villagers were unfamiliar with the temporary roads. This has been greatly improved along the project progress.
As the construction site was far from where the villagers live and the main traffic system, the probability of this risk was low. As both sides of the river were surrounded by cultivated land, the construction site did not occupy the main traffic routes, though some large construction machinery may influence the convenience and safety of the public traffic.

(3) The production impacts
The production impacts included the impact on agricultural production of the peasants and the industrial production of the enterprises. Except occupying some cultivated lands, the project had low impacts on agricultural production. There were some influences on irrigation, which could be minimized with reasonable corresponding plan.
However, the impact on industrial production was much higher. For most enterprises, the occupation of the operation site means the factory had to be shut down, which presented serious financial matter. Similarly, for those companies, such as the resource recovery company, the negative impacts of the project on water quality affected its production.
As the land was occupied, the probability of the risk was very high. During the interviews and discussion forum, the owners of these enterprises claimed that they were willing to accept the decrease of the production capacity, but perceived it unacceptable if the factory had to be shut down. Due to the long construction schedule, the industrial production was affected to a large extent which was linked to not only the survival of the factory but also the income of the residents and the economic development of the region. Therefore the impact of this risk was severe.

(4) Other impacts during construction period
Other impacts in construction period included construction safety, the management of migrant labor, etc. Large construction machine and the stacking of the construction materials presented safety risks to the villagers' daily life. Moreover, the management of the construction personnel, especially the migrant workers, affected the social stability and safety of the local community. However, the probability of these risks was low and may not influence the life of the villagers directly with severe impacts, according to interviews with related stakeholders and discussion forum.

4.3.4. Other social risks

Apart from the risks discussed in the previous section, other social risks associated with this project were identified as a result of interviews and discussion forums.

(1) Changes to the living environment
The living style of the local residents would be, to some extent, affected with the change of landscape, the production mode, the production place, etc. The land acquisition and demolition would lead to the migration of some residents and change the neighborhood relationships. However, the proposal and project plan showed that the residents did not need to move to other regions far from the original place. Therefore, it did not lead to significant change to the living environment. During the interviews and discussion forums, villagers thought they could adapt to the new environment with a reasonable period of time. Moreover, villages commented that they were aware of the long term benefits of the project through the redevelopment of the town and village. Therefore, the risk probability was low.

(2) Concerns on the living standard
As the project needed to occupy a large quantity of cultivated land and shutdown of some factories, local residents faced enormous pressure as their income was affected to a large extent. However, they were reasonably happy during the interviews and discussion forum due to the awareness of public benefits of the project from a long term perspective and the efforts of government to implement a fair compensation plan. Therefore, the risk probability was very low.

(3) Social conflicts
The interviews and discussion forum results showed that the common sources of social conflicts included: dissatisfaction with the compensation scheme, lack of transparency and project decisions, lack of knowledge of the project, lack of awareness of the benefits and impacts of the project from a long term perspective. This social risk could be managed via coordination between government and related stakeholder groups. In addition, the media played a crucial role to clarify misunderstandings in order that the public and the local community would support the project.

4.4. Social risk assessment

According to the above risk identification results, the social risk factors of this project and associated likelihood and consequence are listed in Table 2. The risk value R was calculated as follows:

$$R = \text{Consequence of risk} \times \text{Likelihood of risk}. \quad (1)$$

As shown in Fig. 3, Level I and Level II are defined as the mild risks, Level III and Level IV are defined as the moderate risks, and Level V is the most critical risks.

Table 2
Social risk assessment of Yanshan Trunk River.

No.	Risk factors	Likelihood	Consequence	Final result
R01	Legitimacy and rationality of the projects challenged	Low	Heavy loss	Level III
R021	Unfair land compensation	High	Heavy loss	Level V
R022	Unfair house demolition compensation	High	Heavy loss	Level V
R023	Enterprises losses caused by land acquisition and demolition	High	Heavy loss	Level V
R031	Environmental pollution	High	Moderate loss	Level IV
R032	Construction impact on the residents	General	Moderate loss	Level III
R033	Construction impact on the enterprises	High	Heavy loss	Level V
R034	Transport impact	Low	Mild loss	Level I
R035	Construction safety and management problems	General	Mild loss	Level II
R041	Adapt to the new environment	Low	Mild loss	Level I
R042	Concerns on the standard of living	Low	Mild loss	Level I
R043	Social conflicts	General	Moderate loss	Level III

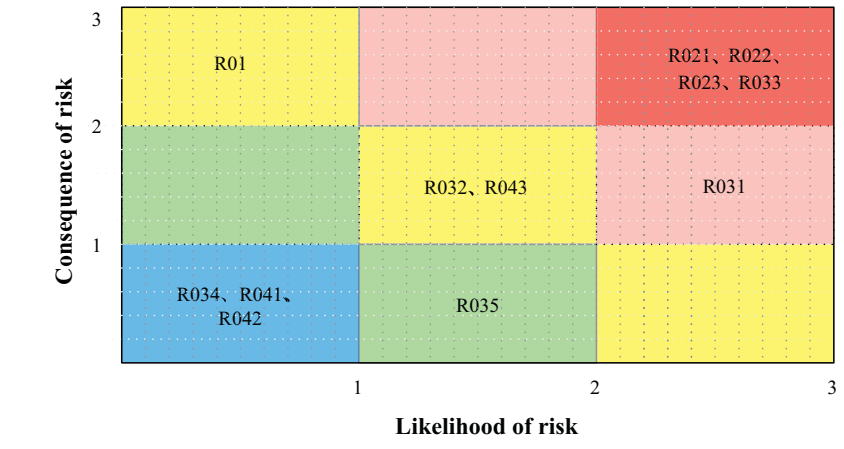
The assessment process started from the collection of comments from the experts, including government officials, project management professionals and social experts. Subsequently, an experts' meeting was held to discuss the likelihood and consequence of each risk. In addition, the result of the investigation to the stakeholder groups and the discussion forum was referred during the meeting. Finally, four social risk factors were ranked most critical for this project, i.e. unfair land compensation (R021), unfair house demolition compensation (R022), enterprises losses caused by land acquisition and demolition (R023), and construction impact on the enterprises (R033).

4.5. Response plan

In order to manage the social risks of this project, the research team was commissioned by the project management office of this case project to organize meetings with experts in order to develop a response plan with regard to the prevention, mitigation and controlling of social risks. Experts were presented the results of social risk assessment, including the likelihood of occurrence and the consequence of each individual social risk. The project files and relevant documents on the request of the stakeholder groups were provided as key references to draft the response plan. Experts were asked to give suggestions on the countermeasures to deal with the social risks, particularly for those with high risks (Level V). Other measures included the prevention and early warning plan, and the general response measures. The principle of this response plan was to maintain social stability and harmony. The development of response plan was an iteration process where several rounds were conducted. The research team worked closely with the project management office to monitor the effectiveness of the response plan. Necessary changes were made to the early versions of the response plan.

4.5.1. Prevention and early warning

As a result of intensive discussion, the prevention and early warning measures were summarized as below:



Note:
Consequence of risk: 1-mild loss, 2-moderate loss, 3-heavy loss
Likelihood of risk: 1-less, 2-general, 3-high
Level I
Level II
Level III
Level IV
Level V

Fig. 3. Social risk assessment of Yanshan Trunk River.

(1) It is imperative to integrate the maintenance of social stability into the agenda of project management to allow monitoring the status of social stability in a regular basis. At the same time, the project management office should consider and examine the various new situations through related stakeholder groups and analyze major issues that may arise
(2) It is necessary to effectively implement the responsibility system for maintaining social stability, clarifying key positions, carrying out target management for safeguarding of social stability and conducting regular evaluation of the performance of related departments. Accountability of those personnel mishandling social risks would be sought for their negligence causing social incidents and serious consequences.
(3) Regular visit to the surrounding areas. This work must shift from routine of passive mode to proactive mode. The project management staff should approach the construction site and community to receive the opinions from related stakeholder groups so that any potential social issues can be resolved in the cradle.
(4) Adhere to the mechanism of regular communications and forecasting system. For the problems commonly raised by stakeholder groups, related staff should investigate and develop practical solutions. As a result, early sign of social risk events could be identified and resolved promptly.
(5) Maintain the effectiveness of communication channel. Once the report of the incident is received, emergency contact number should be able to reach 24 h a day. The staff on duty is responsible for communicating related information to all stakeholder groups. Transparency should be maintained by means of various publicity channels in order to obtain recognition and support from public and local community.

4.5.2. General response plan

Once the social risk events occur, the related departments should trigger the response plan and launch the procedure as the following:

(1) The related departments of the project management office should take the social incidents seriously and send staff to the site immediately. In addition, the communication with the related stakeholder groups should be timely to avoid the spread of conflicts.
(2) Hold the social stability safeguarding meeting as soon as possible, report the situation of the incidents and the response measures, and analyze the effectiveness of these measures.
(3) For those complicated and large-scale social incidents, related department's heads of the project management office need to arrive at the site as soon as possible to mitigate risk and determine solutions.
(4) Take care of any safety related risks to the surrounding community with necessary assistance from police and national security authorities.

4.5.3. Specific response plan for high risk (level V) factors

According to the result of social risk assessment, risks R21 and R22 derived from land acquisition and housing demolition were identified as most critical risks with high likelihood of occurrence and severe consequence. To prevent these risks, the public and local communities need to be consulted in order to minimize the social instability risks. The villager and enterprise owners' benefits should be taken into account, and genuine efforts should be made to minimize economic loss and disturbance to local residents.

To deal with the social risks associated with enterprises (R23 and R33), the construction schedule should be optimized. In addition, the compensation scheme should be fair so as to secure the support of the related stakeholder groups. This project involved relocation of a number of local enterprises. The demolition should be planned rhythmically. For example, those enterprises close to the river with low daily economic outputs could be relocated earlier. From the timing perspective, the demolition work could begin at the construction phase, and it was unnecessary to be earlier than the design phase. Furthermore, most villagers and enterprises preferred land compensation than monetary indemnity. Therefore, the resistance and difficulties of the land acquisition and demolition would be minimized with a combined type of compensation scheme in place.

4.6. Result

Owing to the pro-active response plan, the social risks of the project were successfully controlled. From project initialization, design to construction, there were no social conflicts and serious negative issues which prevented the project from moving forward. The implementation work of the project was progressing according to the schedule. In addition, with more parts of the river being renovated, the landscape looks more beautiful. The positive aspects of the projects for urbanization were gradually revealed which enhanced the acceptance of the social stakeholder groups.

5. Lessons learnt

Based on the case study of Yanshan Trunk River Renovation project, several key points need to be addressed in order to successfully manage the social risks. In addition to establish a sound and systematic risk management system presented in part 4, the management of social risks needs to play more focus on sufficient communication, timely response, effective trust building and the enhancement of social benefits.

(1) *Sufficient communication*. An effective and efficient communication mechanism is crucial for minimizing social risks as the potential conflicts can be identified from related stakeholder groups from very early on. The prerequisite of successful social risk management is to catch the "voice of stakeholders" timely and clearly. It is detrimental to the infrastructure project if any misunderstanding exists about the needs of stakeholders. Sufficient

time should be dedicated on communication. It is not adequate by simply asking the stakeholders to attend a meeting presenting their needs. The communication and engagement of stakeholders is an iterative process which usually needs several times of face to face individual interviews and discussions. For example, one villager initially refused to move and did not tell the reason. After several times of communication, the risk management team found that the reason was that he would possibly lose his job if he moved to a place far away. Therefore, if there were any solutions for him to find a new and better job, he would move. In addition, it is necessary to keep related project information, both pros and cons, transparent to the social stakeholders. It is not appropriate to hold off information as it may make stakeholders nervous which may be exaggerated in infrastructure developments.

(2) *Timely response*. Corresponding to any emerging social risks, timely response or even "agile response" is very important. Once the risk events are emerging or just occurred, the project management office must initiate the response plans immediately without any delay. It is necessary to set up the organizational structure of social stability safeguarding department with clearly defined responsibilities and efficient working mechanism. All staff of this organization can easily find their own position when risk events occurred and control the situation in a timely and orderly manner. In addition, the headquarter should delegate authorities to lower level units to eliminate the social risks and control the site in the first instance. Another important point to ensure timely response is that the response plan should be sufficiently prepared in advance, particularly for high risk factors. As one expert pointed out: "… we must carefully prepare the countermeasures to deal with the risks and try not to miss anything even if there is no signs that they are coming… to be a pessimist and always bearing risks in mind is much better than to be an optimist with fluke mind."

(3) *Effective trust building*. The management of social risk management is to build trust to the stakeholders. Social conflicts usually come from distrust. Therefore, it is imperative to recognize the social stakeholders and to respect their feeling. Making commitment to bring value to the stakeholders and keeping promise all the time is a good way to build trust and prevent potential social risks. In addition, if there are any unreasonable requirements from some stakeholders, do not escalate the commitment which may not be realized. Otherwise it will lead to "a vicious circle" of distrust. If such a scenario does occur, it is better to emphasize the "public first" principle and the risk management team should coordinate major decisions with a consideration of all stakeholder groups' interests. When conflicts exist between the public interest and private interests, the public interest should be given priority. There is no doubt that the basic objective of major policy decisions is to benefit the society, and to meet the interests and aspirations of the vast majority of community. Therefore, a "public first" principle should be established during the process of risk management (Bai, 2008). Being fair to all the stakeholders is also crucial to build trust.

(4) *Enhancement of social benefits*. Social risk management is not only the prevention of social conflicts but also the enhancement of common social benefits. Indeed, the level of awareness of long term benefits of project was found very influential to the social acceptance of the local community in this project. As one villager put, "…if my hometown can be rebuilt more beautiful and bring more opportunities on increasing income, I will be more willing to move out the village temporary with high expectation." Therefore, efforts are needed to convey these benefits and knowledge to the public and community. The media plays a crucial role to achieve this goal. Media may either promote risk communication or disturb public awareness or psychological state (Xie and Zheng, 2003). In fact, the project can drive the development of the local economy, enjoy appreciation of rim land, increase the opportunity of acquiring wealth, and improve the welfare for the peasant in long term, despite some inconvenience and some economic loss in short term. Therefore, media should convey this information to related stakeholders and the public in an accurate manner.

Apart from traditional functions such as water conservation, flood control and ecological management, the hydraulic infrastructure project can also offer more functions to meet the high level demands of the local residents and enterprises. For example, if the land use meets the conditions, the project can build some landscapes along the side of the river such as parks, recreational facilities, and tourism facilities to improve the quality of life of local residents. Landscape channel and riverway should be reserved in the master plan and bridges should be built in suitable locations to provide a traffic connection linking both sides of the river.

It is worth noting that some factors reported in previous studies are not covered in the social risk assessment in this case as they are not suitable for hydraulic infrastructure projects however may be suitable for other types of construction projects. For example, the location selection can also be a social risk in some engineering projects (Zhu, 2012). However for the hydraulic infrastructure project, there are limited choices for the location. Therefore, this factor was deleted from the list. For other types of infrastructure projects, the list of social risk factors should be adjusted with a consideration of their unique characteristics.

6. Conclusion

The social risk management is emerging as a new measure of the social administration in China. There are numerous benefits associated with establishing social risk management system of major infrastructure projects such as discovering potential risks timely, innovative social risk management, minimizing the occurrence of social conflicts, ensuring the

smooth implementation of major projects, and maintaining social stability.

This paper proposed a social risk management system using a case study approach through a detailed description and critical analysis of the experience of related practices in the Yanshan Trunk River Renovation project. This system includes forming a social risk management group, understanding the progress of the project, recognizing stakeholders, and adopting multiple information collection approaches such as observations, meeting, interviews and discussion forums to identify and assess social risks, developing response plan and monitoring risks.

Through this case study, firstly, 12 main social risk factors related to this hydraulic infrastructure project were identified. Secondly, the level of each social risk was assessed. It was found that unfair land compensation, unfair house demolition compensation, enterprise losses caused by land acquisition and demolition, and construction impacts on the enterprises presented as the most critical social risks associated with the studied case. In addition, a response plan was proposed together with recommendations to deal with social risks existed in key infrastructure projects. These strategies place focus on preventing social risks in the first instance and recognizing the role of mitigation and coping mechanism once the events occurred. This study also found that the local community was more willing to support the project if they were more aware of the long term benefits of the project and the transparency of project decision was maintained. This indicates that social aspects of infrastructure developments should not be overlooked and focus should be placed on the education of local community of the long term tangible benefits of infrastructure developments. In addition, it was revealed that sufficient communication, timely response, effective trust building and the enhancement of social benefits are extremely important measures for successful management of social risks. Although the findings of this case study are probably not enough to build ground theory on social risk management due to its inherent single settings, they may provide a useful reference to the social risk management of future hydraulic infrastructure projects in China and could supplement the traditional body of knowledge on project risk management. For instance, the systematic approach proposed in this study could be adopted in the study of social risk management in other infrastructure projects. The identified social risks together with the structure can be served as a useful starting point of managing social risks in infrastructure developments. It is worth noting that contextual factors (e.g. location, culture, local economy status) should be taken into consideration when applying this framework in different contexts.

In the future research, multiple case studies can be employed to validate or extend the findings from this study which is based on a single case, e.g., whether the 12 risk factors cover all the social risks of an infrastructure project; whether there are any other social risks during the development and construction of an infrastructure project; what are the social risk factors in the entire life-cycle of an infrastructure project; and whether there exist any other important steps or links in a social risk management system. Similarly, meta-data investigation and analysis can be utilized to examine the concise measures of logic coherence, causal relationship between risk countermeasures and social risk factors, e.g., how the detail measures of trust building affect each social risk factor, and explore the correlations amongst various risk factors.

Conflict of interest

There are no conflicts.

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